

Task4: Microservices with Kafka Messaging Queue

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GitHub: https://github.com/mariiaivanchenko/Software-Architecture/tree/micro_mq

Завдання базується на функціоналі розробленому у попередньому завданні, і є його подальшим розвитком.

До **messages-service** необхідно додати:

- у якості каналу доставки повідомлень між **facade-service** та **messages-service** використати message queue
- можливість запускати одночасно декілька копій **messages-service**
- **facade-service** випадковим чином обирає до якої копії **messages-service** звертатись для читання повідомлень

Я вирішила використовувати **Kafka messages queue**.

Функціональність системи

У якості клієнта цього разу я використовувала файл *requests.http*, де за допомогою розширення *REST Client*, можна легко надсилати запити.

```
requests.http > POST /facade-service
2 references
1  @FACADE_URL = http://127.0.0.1:5000/facade-service
2
3  ### POST request for facade-service
4  Send Request
5  POST {{FACADE_URL}}
6  Content-Type: text/plain
7
8  msg02
9
10 ### GET request for facade-service
11 Send Request
12 GET {{FACADE_URL}}
```

Запуск елементів

Для запуску усіх необхідних елементів:

- 1 *facade-service*, 2 *messages-service*, 1 *config-server*
- 3 *logging-service*, 3 *Hazelcast cluster nodes*
- opened **Docker Desktop** + *docker-compose.yml* + *kafka*

```
docker pull apache/kafka
```

```
➤ Lab4 docker pull apache/kafka
Using default tag: latest
latest: Pulling from apache/kafka
f18232174bc9: Pull complete
c3f73af89931: Pull complete
729fc64ae8c1: Pull complete
28bd55152645: Pull complete
2fai1f65d07a3: Pull complete
b42f722acfd6: Pull complete
73851e29d6a7: Pull complete
b8eeb529f1af: Pull complete
a722193c2f26: Pull complete
643bf8a7c247: Pull complete
35e384b206f: Pull complete
Digest: sha256:3770939115c4b072e9cee9369d08bd69712fde55f9902f46d793f64848dedc75
Status: Downloaded newer image for apache/kafka:latest
docker.io/apache/kafka:latest

What's Next?
View a summary of image vulnerabilities and recommendations → docker scout quickview apache/kafka
➤ Lab4
```

використовувала скрипт *start.sh*, який запускає усі відповідні сервіси, запускає **Docker**, створює відповідний **topic** та витягує змінні з *configuration.ini*, відповідний скрипт вбиває усі процеси *stop.sh*.

```
○ .venv➔ Lab4 ./start.sh
Loading configuration file data...
[+] Building 0.0s (0/0)
[+] Running 5/5
✓ Network lab4_default Created
✓ Container controller-1 Started
✓ Container broker-2 Started
✓ Container broker-3 Started
✓ Container broker-1 Started
```

```
Created topic messages-topic.
Starting hazelcast nodes...
#####
# JAVA=/usr/bin/java
```

```
Members {size:3, ver:3} [
  Member [127.0.0.1]:5701 - c17f8837-b9c6-4ae3-bf37-f333df702686
  Member [127.0.0.1]:5702 - 359187f2-2a27-486c-ae81-da648557827e
  Member [127.0.0.1]:5703 - b7155b88-ee27-4723-83fb-70deb390b869 this
]
```

```
INFO: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
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WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
```

Надсилаємо повідомлення

спершу **GET**, перевірити, що повідомлень ще не надходило.

```
messages-service.py x Response(2418ms) x
1 HTTP/...
2 Server: Werkzeug/3.1.3 Python/3.13.2
3 Date: Fri, 25 Apr 2025 19:58:23 GMT
4 Content-Type: text/html; charset=utf-8
5 Content-Length: 38
6 Connection: close
7
8 Logging response:
9 Messages response:
```

Коли надходять запити на **facade-service**:

- якщо це **POST**, додаємо повідомлення до черги у ролі **Producer**

кожен з відповідних **messages-service** вичитує повідомлення та зберігає їх.

- якщо це **GET**, також робимо запит на **config-server**, але тепер для адрес для **messages-service**. надсилаємо запит на обраний сервіс, який через **HTTP** повертає збережені повідомлення.

msg1 - <http://127.0.0.1:5003/logging-service>

```
Chooesd logging in POST: http://127.0.0.1:5003/logging-service
Recieved messages for logging: msg1
127.0.0.1 - - [25/Apr/2025 23:07:44] "POST /logging-service HTTP/1.1" 200 -
Sent to message queue: {'uuid': 'ca9e10a5-ed2c-4c52-8173-500b4dc2110b', 'message': 'msg1'}
127.0.0.1 - - [25/Apr/2025 23:07:44] "POST /facade-service HTTP/1.1" 200 -
Received from messages queue (service 2): {'uuid': 'ca9e10a5-ed2c-4c52-8173-500b4dc2110b', 'message': 'msg1'}
Received from messages queue (service 1): {'uuid': 'ca9e10a5-ed2c-4c52-8173-500b4dc2110b', 'message': 'msg1'}
```

msg2 - <http://127.0.0.1:5003/logging-service>

```
Chooesd logging in POST: http://127.0.0.1:5003/logging-service
Recieved messages for logging: msg2
127.0.0.1 - - [25/Apr/2025 23:08:55] "POST /logging-service HTTP/1.1" 200 -
Sent to message queue: {'uuid': 'ce71a353-a222-4513-b30a-88fc6f94474c', 'message': 'msg2'}
127.0.0.1 - - [25/Apr/2025 23:08:55] "POST /facade-service HTTP/1.1" 200 -
Received from messages queue (service 1): {'uuid': 'ce71a353-a222-4513-b30a-88fc6f94474c', 'message': 'msg2'}
Received from messages queue (service 2): {'uuid': 'ce71a353-a222-4513-b30a-88fc6f94474c', 'message': 'msg2'}
```

msg3 - <http://127.0.0.1:5001/logging-service>

```
Chooesd logging in POST: http://127.0.0.1:5001/logging-service
Recievd messages for logging: msg3
127.0.0.1 - - [25/Apr/2025 23:09:11] "POST /logging-service HTTP/1.1" 200 -
Sent to message queue: {'uuid': '724a9b44-f5e5-4290-b21b-1b782ef6c75a', 'message': 'msg3'}
127.0.0.1 - - [25/Apr/2025 23:09:11] "POST /facade-service HTTP/1.1" 200 -
Received from messages queue (service 1): {'uuid': '724a9b44-f5e5-4290-b21b-1b782ef6c75a', 'message': 'msg3'}
Received from messages queue (service 2): {'uuid': '724a9b44-f5e5-4290-b21b-1b782ef6c75a', 'message': 'msg3'}
□
```

msg4 - <http://127.0.0.1:5003/logging-service>

```
Chooesd logging in POST: http://127.0.0.1:5003/logging-service
Recievd messages for logging: msg4
127.0.0.1 - - [25/Apr/2025 23:09:24] "POST /logging-service HTTP/1.1" 200 -
Sent to message queue: {'uuid': '143c4c93-840d-42f8-90d2-3db9bb33aba1', 'message': 'msg4'}
127.0.0.1 - - [25/Apr/2025 23:09:24] "POST /facade-service HTTP/1.1" 200 -
Received from messages queue (service 2): {'uuid': '143c4c93-840d-42f8-90d2-3db9bb33aba1', 'message': 'msg4'}
Received from messages queue (service 1): {'uuid': '143c4c93-840d-42f8-90d2-3db9bb33aba1', 'message': 'msg4'}
□
```

msg5- <http://127.0.0.1:5001/logging-service>

```
Chooesd logging in POST: http://127.0.0.1:5001/logging-service
Recievd messages for logging: msg5
127.0.0.1 - - [25/Apr/2025 23:09:43] "POST /logging-service HTTP/1.1" 200 -
Sent to message queue: {'uuid': 'b7c18f7a-f5d6-4666-b1c0-8aee08de029e', 'message': 'msg5'}
127.0.0.1 - - [25/Apr/2025 23:09:43] "POST /facade-service HTTP/1.1" 200 -
Received from messages queue (service 2): {'uuid': 'b7c18f7a-f5d6-4666-b1c0-8aee08de029e', 'message': 'msg5'}
Received from messages queue (service 1): {'uuid': 'b7c18f7a-f5d6-4666-b1c0-8aee08de029e', 'message': 'msg5'}
□
```

msg6 - <http://127.0.0.1:5002/logging-service>

```
Chooesd logging in POST: http://127.0.0.1:5002/logging-service
Recievd messages for logging: msg6
127.0.0.1 - - [25/Apr/2025 23:09:56] "POST /logging-service HTTP/1.1" 200 -
Sent to message queue: {'uuid': '0d9199f5-1547-46e4-b99f-cb73a0a83099', 'message': 'msg6'}
127.0.0.1 - - [25/Apr/2025 23:09:56] "POST /facade-service HTTP/1.1" 200 -
Received from messages queue (service 2): {'uuid': '0d9199f5-1547-46e4-b99f-cb73a0a83099', 'message': 'msg6'}
Received from messages queue (service 1): {'uuid': '0d9199f5-1547-46e4-b99f-cb73a0a83099', 'message': 'msg6'}
□
```

msg7 - <http://127.0.0.1:5001/logging-service>

```
Chooed logging in POST: http://127.0.0.1:5001/logging-service
Recieved messages for logging: msg7
127.0.0.1 - - [25/Apr/2025 23:10:07] "POST /logging-service HTTP/1.1" 200 -
Sent to message queue: {'uuid': '7200574c-ae19-44c0-b3db-17b973d47015', 'message': 'msg7'}
127.0.0.1 - - [25/Apr/2025 23:10:07] "POST /facade-service HTTP/1.1" 200 -
Received from messages queue (service 2): {'uuid': '7200574c-ae19-44c0-b3db-17b973d47015', 'message': 'msg7'}
Received from messages queue (service 1): {'uuid': '7200574c-ae19-44c0-b3db-17b973d47015', 'message': 'msg7'}
```

msg8 - <http://127.0.0.1:5002/logging-service>

```
Chooed logging in POST: http://127.0.0.1:5002/logging-service
Recieved messages for logging: msg8
127.0.0.1 - - [25/Apr/2025 23:10:19] "POST /logging-service HTTP/1.1" 200 -
Sent to message queue: {'uuid': '481889e0-1746-4e69-b085-966b4012041a', 'message': 'msg8'}
Received from messages queue (service 1): {'uuid': '481889e0-1746-4e69-b085-966b4012041a', 'message': 'msg8'}
Received from messages queue (service 2): {'uuid': '481889e0-1746-4e69-b085-966b4012041a', 'message': 'msg8'}
127.0.0.1 - - [25/Apr/2025 23:10:19] "POST /facade-service HTTP/1.1" 200 -
```

msg9 - <http://127.0.0.1:5003/logging-service>

```
Chooed logging in POST: http://127.0.0.1:5003/logging-service
Recieved messages for logging: msg9
127.0.0.1 - - [25/Apr/2025 23:11:42] "POST /logging-service HTTP/1.1" 200 -
Sent to message queue: {'uuid': 'b9bfb564-19d7-474c-ae1a-2ec86e04b2c2', 'message': 'msg9'}
127.0.0.1 - - [25/Apr/2025 23:11:43] "POST /facade-service HTTP/1.1" 200 -
Received from messages queue (service 1): {'uuid': 'b9bfb564-19d7-474c-ae1a-2ec86e04b2c2', 'message': 'msg9'}
Received from messages queue (service 2): {'uuid': 'b9bfb564-19d7-474c-ae1a-2ec86e04b2c2', 'message': 'msg9'}
```

msg10 - <http://127.0.0.1:5003/logging-service>

```
Chooed logging in POST: http://127.0.0.1:5003/logging-service
Recieved messages for logging: msg10
127.0.0.1 - - [25/Apr/2025 23:12:07] "POST /logging-service HTTP/1.1" 200 -
Sent to message queue: {'uuid': '44388598-d5e7-4982-83ad-ad95b2dcb39e', 'message': 'msg10'}
127.0.0.1 - - [25/Apr/2025 23:12:07] "POST /facade-service HTTP/1.1" 200 -
Received from messages queue (service 2): {'uuid': '44388598-d5e7-4982-83ad-ad95b2dcb39e', 'message': 'msg10'}
Received from messages queue (service 1): {'uuid': '44388598-d5e7-4982-83ad-ad95b2dcb39e', 'message': 'msg10'}
```

Перевірка з GET запитами

```
Chooed logging in GET: http://127.0.0.1:5001/logging-service
Chooed messages in GET: http://127.0.0.1:5005/messages-service
127.0.0.1 - - [25/Apr/2025 23:12:38] "GET /logging-service HTTP/1.1" 200 -
127.0.0.1 - - [25/Apr/2025 23:12:39] "GET /messages-service HTTP/1.1" 200 -
127.0.0.1 - - [25/Apr/2025 23:12:39] "GET /facade-service HTTP/1.1" 200 -
```

```
Response(631ms) ×
1 HTTP/1.1 200 OK
2 Server: Werkzeug/3.1.3 Python/3.13.2
3 Date: Fri, 25 Apr 2025 20:12:39 GMT
4 Content-Type: text/html; charset=utf-8
5 Content-Length: 156
6 Connection: close
7
8 Logging response: msg9, msg3, msg4, msg8, msg7, msg6, msg2, msg5, msg1, msg10
9 Messages response: msg1, msg2, msg3, msg4, msg5, msg6, msg7, msg8, msg9, msg10
```

```
Chooesd logging in GET: http://127.0.0.1:5001/logging-service
Chooesd messages in GET: http://127.0.0.1:5004/messages-service
127.0.0.1 - - [25/Apr/2025 23:13:41] "GET /logging-service HTTP/1.1" 200 -
127.0.0.1 - - [25/Apr/2025 23:13:41] "GET /messages-service HTTP/1.1" 200 -
127.0.0.1 - - [25/Apr/2025 23:13:41] "GET /facade-service HTTP/1.1" 200 -
□
```

```
Response(501ms) ×
1 HTTP/1.1 200 OK
2 Server: Werkzeug/3.1.3 Python/3.13.2
3 Date: Fri, 25 Apr 2025 20:13:41 GMT
4 Content-Type: text/html; charset=utf-8
5 Content-Length: 156
6 Connection: close
7
8 Logging response: msg9, msg3, msg4, msg8, msg7, msg6, msg2, msg5, msg1, msg10
9 Messages response: msg1, msg2, msg3, msg4, msg5, msg6, msg7, msg8, msg9, msg10
```

```
Chooesd logging in GET: http://127.0.0.1:5002/logging-service
Chooesd messages in GET: http://127.0.0.1:5005/messages-service
127.0.0.1 - - [25/Apr/2025 23:14:15] "GET /logging-service HTTP/1.1" 200 -
127.0.0.1 - - [25/Apr/2025 23:14:15] "GET /messages-service HTTP/1.1" 200 -
127.0.0.1 - - [25/Apr/2025 23:14:15] "GET /facade-service HTTP/1.1" 200 -
□
```

```
Response(142ms) X
1 HTTP/1.1 200 OK
2 Server: Werkzeug/3.1.3 Python/3.13.2
3 Date: Fri, 25 Apr 2025 20:14:15 GMT
4 Content-Type: text/html; charset=utf-8
5 Content-Length: 156
6 Connection: close
7
8 Logging response: msg9, msg3, msg4, msg8, msg7, msg6, msg2, msg5, msg1, msg10
9 Messages response: msg1, msg2, msg3, msg4, msg5, msg6, msg7, msg8, msg9, msg10
```

Перевірка реплікації

спершу не запускаю **messages-service** у **start.sh.**, щоб повідомлення тимчасово не вичитувались та зберігались у **message queue**.

```
42 # messages
43 # echo "Starting messages services..."
44 # for i in `seq 0 1`;
45 # do
46 #     port=$((MESSAGES_START_PORT + $i))
47 #     flask --app messages-service run --port=$port &
48 #     sleep 3
49 # done
50
51 # config
52 flask --app config-server run --port=$PORTS_CONFIG_PORT &
53
```

надсилаю 10 повідомлень.

```
Chooed logging in POST: http://127.0.0.1:5002/logging-service
Recieved messages for logging: msg1
127.0.0.1 - - [25/Apr/2025 23:37:42] "POST /logging-service HTTP/1.1" 200 -
Sent to message queue: {'uid': '03d5764c-880c-4c7b-b6f6-91583b3fad55', 'message': 'msg1'}
127.0.0.1 - - [25/Apr/2025 23:37:42] "POST /facade-service HTTP/1.1" 200 -
127.0.0.1 - - [25/Apr/2025 23:37:47] "GET /config-server/service.names/logging HTTP/1.1" 200 -
127.0.0.1 - - [25/Apr/2025 23:37:48] "GET /logging-service HTTP/1.1" 200 -
Chooed logging in POST: http://127.0.0.1:5001/logging-service
Recieved messages for logging: msg2
127.0.0.1 - - [25/Apr/2025 23:37:48] "POST /logging-service HTTP/1.1" 200 -
Sent to message queue: {'uid': 'd726911e-b7d3-47d6-a13a-b30f8e8557fa', 'message': 'msg2'}
127.0.0.1 - - [25/Apr/2025 23:37:48] "POST /facade-service HTTP/1.1" 200 -
127.0.0.1 - - [25/Apr/2025 23:37:51] "GET /config-server/service.names/logging HTTP/1.1" 200 -
127.0.0.1 - - [25/Apr/2025 23:37:51] "GET /logging-service HTTP/1.1" 200 -
Chooed logging in POST: http://127.0.0.1:5001/logging-service
Recieved messages for logging: msg3
127.0.0.1 - - [25/Apr/2025 23:37:51] "POST /logging-service HTTP/1.1" 200 -
Sent to message queue: {'uid': '77a57a00-86d1-4293-ab53-9e44876453a6', 'message': 'msg3'}
127.0.0.1 - - [25/Apr/2025 23:37:51] "POST /facade-service HTTP/1.1" 200 -
```

визначити Leader та вимкнути його.

```
docker exec -it broker-1 /opt/kafka/bin/kafka-topics.sh --bootstrap-server broke
docker logs broker-1
docker stop broker-1
```



```
.venv> Lab4 docker exec -it broker-1 /opt/kafka/bin/kafka-topics.sh --bootstrap-server broker-1:19092,broker-2:19092,broker-3:19092
--describe --topic messages-topic
Topic: messages-topic TopicId: W93oB0eqRqCa_jxyXawGPw PartitionCount: 1 ReplicationFactor: 3 Configs:
Topic: messages-topic Partition: 0 Leader: 2 Replicas: 2,3,4 Isr: 2,3,4 Elr: LastKnownElr:
.venv> Lab4 docker logs broker-1
====> User
uid=1000(appuser) gid=1000(appuser) groups=1000(appuser)
====> Setting default values of environment variables if not already set.
CLUSTER_ID not set. Setting it to default value: "5L6g3nShT-eMctK--X86sw"
====> Configuring ...
Running in KRaft mode...
====> Launching ...
====> Using provided cluster id 5L6g3nShT-eMctK--X86sw ...
[2025-04-25 20:56:25,180] INFO Registered kafka:type=kafka.Log4jController MBean (kafka.utils.Log4jControllerRegistration$)
[2025-04-25 20:56:27,055] INFO Registered signal handlers for TERM, INT, HUP (org.apache.kafka.common.utils.LoggingSignalHandler)
[2025-04-25 20:56:27,068] INFO [BrokerServer id=2] Transition from SHUTDOWN to STARTING (kafka.server.BrokerServer)
```

запускаю 2 **messages-service**, кожен з них отримав повідомлення.

```
flask --app messages-service run --port=5004 &
flask --app messages-service run --port=5005 &
```

```
.venv> Lab4 flask --app messages-service run --port=5004 &
[1] 7698
.venv> Lab4 * Serving Flask app 'messages-service'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5004
Press CTRL+C to quit
Received from messages queue (service 1): {'uid': '6eac24ea-a410-4ff7-847b-513e72007e8', 'message': 'msg1'}
Received from messages queue (service 1): {'uid': '637f62fd-a1e6-46bd-b2d3-251b2f50de29', 'message': 'msg2'}
Received from messages queue (service 1): {'uid': 'c757893e-b43a-435c-a228-5562e354250b', 'message': 'msg3'}
Received from messages queue (service 1): {'uid': 'a37b6daa-b559-438e-9947-003fd118ad78', 'message': 'msg4'}
Received from messages queue (service 1): {'uid': 'b11ae091-1af9-4528-b2c4-066b0b837f1', 'message': 'msg5'}
Received from messages queue (service 1): {'uid': 'd955b079-81f4-4268-9925-24d20ebf723c', 'message': 'msg6'}
Received from messages queue (service 1): {'uid': 'd511d3c5-58c5-4c89-af27-28124cfff96e', 'message': 'msg7'}
Received from messages queue (service 1): {'uid': 'e9ea3ca3-788c-49ea-a989-1f3ee86bb07', 'message': 'msg8'}
Received from messages queue (service 1): {'uid': 'd5f68ea2-bae1-45e4-a3b4-ae76b44c1f04', 'message': 'msg9'}
Received from messages queue (service 1): {'uid': 'eab2998a-a8d1-4f0b-8afe-9fa84bc1d525', 'message': 'msg10'}
```

```
.venv> Lab4 flask --app messages-service run --port=5005 &
[1] 77418
.venv> Lab4 * Serving Flask app 'messages-service'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5005
Press CTRL+C to quit
Received from messages queue (service 2): {'uid': '6eac24ea-a410-4ff7-847b-513e72007e8', 'message': 'msg1'}
Received from messages queue (service 2): {'uid': '637f62fd-a1e6-46bd-b2d3-251b2f50de29', 'message': 'msg2'}
Received from messages queue (service 2): {'uid': 'c757893e-b43a-435c-a228-5562e354250b', 'message': 'msg3'}
Received from messages queue (service 2): {'uid': 'a37b6daa-b559-438e-9947-003fd118ad78', 'message': 'msg4'}
Received from messages queue (service 2): {'uid': 'b11ae091-1af9-4528-b2c4-066b0b837f1', 'message': 'msg5'}
Received from messages queue (service 2): {'uid': 'd955b079-81f4-4268-9925-24d20ebf723c', 'message': 'msg6'}
Received from messages queue (service 2): {'uid': 'd511d3c5-58c5-4c89-af27-28124cfff96e', 'message': 'msg7'}
Received from messages queue (service 2): {'uid': 'e9ea3ca3-788c-49ea-a989-1f3ee86bb07', 'message': 'msg8'}
Received from messages queue (service 2): {'uid': 'd5f68ea2-bae1-45e4-a3b4-ae76b44c1f04', 'message': 'msg9'}
Received from messages queue (service 2): {'uid': 'eab2998a-a8d1-4f0b-8afe-9fa84bc1d525', 'message': 'msg10'}
```

перевіряю надіславши **GET**.

```
Response(486ms) ×
1 HTTP/1.1 200 OK
2 Server: Werkzeug/3.1.3 Python/3.13.2
3 Date: Fri, 25 Apr 2025 21:03:58 GMT
4 Content-Type: text/html; charset=utf-8
5 Content-Length: 156
6 Connection: close
7
8 Logging response: msg9, msg1, msg4, msg10, msg6, msg5, msg7, msg2, msg8, msg3
9 Messages response: msg1, msg2, msg3, msg4, msg5, msg6, msg7, msg8, msg9, msg10
```